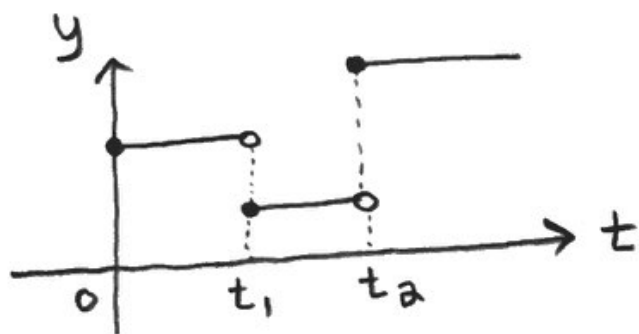


Chapter 7 The Laplace Transform

The Laplace Transform is a powerful tool to solve an IVP of the form:

$$\begin{cases} ay''(t) + by'(t) + cy(t) = f(t) \\ y(0) = y_0 \\ y'(0) = y_1. \end{cases}$$

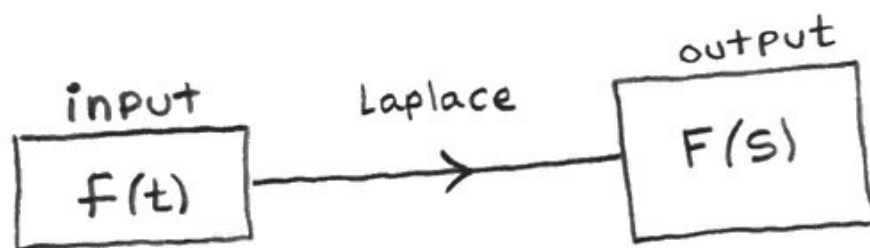
The Laplace transform turns the IVP into an algebraic equation. Laplace transform is very efficient to solve the IVP when $f(t)$ has discontinuities such as:



$$y = f(t).$$

7.1 Definition of the Laplace Transform

The Laplace Transform is an operator that transforms a function $f(t)$ into another function $F(s)$.



$$\mathcal{L}\{f(t)\} = F(s)$$

↓
The Laplace transform
of $f(t)$

Definition The Laplace transform of $f(t)$

is

$$\mathcal{L}\{f(t)\} = \underbrace{\int_0^{\infty} e^{-st} f(t) dt}_{F(s)}.$$

When we compute :

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt,$$

We always assume that s is large enough to guarantee that the integral converges.

ex: ① Find $\mathcal{L}\{1\}$.

$$\mathcal{L}\{1\} = \int_0^{\infty} 1 \cdot e^{-st} dt$$

$$= \left(\frac{e^{-st}}{-s} \right) \Big|_0^{\infty}$$

$$= \lim_{t \rightarrow \infty} \frac{-e^{-st}}{s} + \frac{e^0}{s}$$

Assume that $s > 0$ so that $\lim_{t \rightarrow \infty} e^{-st} = 0$.

$$\Rightarrow \boxed{\mathcal{L}\{1\} = 1/s}$$

ex: ② Derive a formula for

$$\mathcal{L}\{t^n\} \quad \text{for } n=1,2,3,\dots$$

By definition: $\mathcal{L}\{t^n\} = \int_0^{\infty} t^n e^{-st} dt$

Let's use integration by parts :

$$\int u dv = uv - \int v du$$

with $u = t^n$ and $dv = e^{-st} dt$.

$$\Rightarrow du = nt^{n-1} dt \quad \text{and} \quad v = -\frac{e^{-st}}{s}.$$

$$\begin{aligned} \mathcal{L}\{t^n\} &= -\frac{t^n e^{-st}}{s} \Big|_0^{\infty} + \frac{n}{s} \int_0^{\infty} t^{n-1} e^{-st} dt \\ &= \left(\lim_{t \rightarrow \infty} \frac{-t^n}{s e^{st}} \right) + 0 + \frac{n}{s} \int_0^{\infty} t^{n-1} e^{-st} dt \end{aligned}$$

If $s > 0$, then $\lim_{t \rightarrow \infty} \frac{t^n}{e^{st}} = 0$ [Apply l'Hôpital n times]

$$\Rightarrow \mathcal{L}\{t^n\} = \frac{n}{s} \int_0^{\infty} t^{n-1} e^{-st} dt$$

$$\Rightarrow \mathcal{L}\{t^n\} = \frac{n}{s} \mathcal{L}\{t^{n-1}\}$$

Let's look at a few cases for n .

$$n=1 \rightarrow \mathcal{L}\{t\} = \frac{1}{s} \mathcal{L}\{1\} = \frac{1}{s} \left(\frac{1}{s} \right) = \frac{1}{s^2}$$

from example ①, $\mathcal{L}\{1\} = 1/s$

$$n=2 \rightarrow \mathcal{L}\{t^2\} = \frac{2}{s} \mathcal{L}\{t\} = \frac{2}{s} \left(\frac{1}{s^2} \right) = \frac{2}{s^3}$$

$$n=3 \rightarrow \mathcal{L}\{t^3\} = \frac{3}{s} \mathcal{L}\{t^2\} = \frac{3}{s} \left(\frac{2}{s^3} \right) = \frac{6}{s^4}$$

$$n=4 \rightarrow \mathcal{L}\{t^4\} = \frac{4}{s} \mathcal{L}\{t^3\} = \frac{4}{s} \left(\frac{6}{s^4} \right) = \frac{24}{s^5}$$

We see that in general:

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}} \quad \text{for } n=1, 2, 3, \dots$$

The Laplace Transform is a Linear operator.

$$\mathcal{L}\{\alpha f(t) + \beta g(t)\} = \alpha \mathcal{L}\{f(t)\} + \beta \mathcal{L}\{g(t)\}$$

For any constants α and β .

This can be generalized to :

$$\begin{aligned} \mathcal{L}\{c_1 f_1(t) + c_2 f_2(t) + \dots + c_n f_n(t)\} \\ = c_1 \mathcal{L}\{f_1(t)\} + c_2 \mathcal{L}\{f_2(t)\} + \dots + c_n \mathcal{L}\{f_n(t)\} \end{aligned}$$

for any constants $c_1, c_2, \dots, c_n$

The above formula is another example of the Principle of Superposition.

ex: ③ Find $\mathcal{L}\{6t^5 - 2t^3 + 4t + 8\}$.

We already know that $\mathcal{L}\{1\} = 1/s$ and

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}} \text{ for } n=1,2,3,\dots$$

By Linearity of the Laplace Transform,

$$\mathcal{L}\{6t^5 - 2t^3 + 4t + 8\} = 6\mathcal{L}\{t^5\} - 2\mathcal{L}\{t^3\} + 4\mathcal{L}\{t\} + 8\mathcal{L}\{1\}$$

$$= 6\left(\frac{5!}{s^6}\right) - 2\left(\frac{3!}{s^4}\right) + 4\left(\frac{1}{s^2}\right) + 8\left(\frac{1}{s}\right)$$

$$= \boxed{\frac{720}{s^6} - \frac{12}{s^4} + \frac{4}{s^2} + \frac{8}{s}}$$

Note : Doing example ③ from $\int_0^\infty e^{-st} f(t) dt$ would have been tedious. It is much easier to use previously derived formulas and properties.

ex: (4) Find $\mathcal{L}\{e^{at}\}$.

By definition :

$$\mathcal{L}\{e^{at}\} = \int_0^{\infty} e^{-st} e^{at} dt$$

$$= \int_0^{\infty} e^{(a-s)t} dt$$

$$= \left. \frac{e^{(a-s)t}}{(a-s)} \right|_0^{\infty}$$

$$= \lim_{t \rightarrow \infty} \frac{e^{(a-s)t}}{(a-s)} - \frac{e^0}{(a-s)}$$

if $s > a$, then $\lim_{t \rightarrow \infty} e^{(a-s)t} = 0$

$$\Rightarrow \boxed{\mathcal{L}\{e^{at}\} = \frac{1}{s-a}}$$

ex: ⑤ Find $\mathcal{L}\{\cos(\omega t)\}$ and $\mathcal{L}\{\sin(\omega t)\}$.

We could use $\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$

but it is easier to use Euler's Formula.

$$e^{i\theta} = \cos \theta + i \sin(\omega t)$$

$$\Rightarrow e^{i\omega t} = \cos(\omega t) + i \sin(\omega t)$$

$$\mathcal{L}\{e^{i\omega t}\} = \mathcal{L}\{\cos(\omega t)\} + i \mathcal{L}\{\sin(\omega t)\}$$

$$\text{since: } \mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

$$\text{then: } \mathcal{L}\{e^{i\omega t}\} = \frac{1}{s-i\omega}$$

$$\mathcal{L}\{e^{i\omega t}\} = \frac{1}{s-i\omega} \left(\frac{s+i\omega}{s+i\omega} \right)$$

$$= \frac{s+i\omega}{s^2+\omega^2}$$

$$= \frac{s}{s^2+\omega^2} + i \frac{\omega}{s^2+\omega^2}$$

$$\mathcal{L}\{e^{i\omega t}\} = \mathcal{L}\{\cos(\omega t)\} + i \mathcal{L}\{\sin(\omega t)\}$$

By matching the real and imaginary parts
we obtain :

$$\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2+\omega^2}$$

$$\mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2+\omega^2}$$

ex: ⑥ Evaluate $\mathcal{L}\{3e^{2t} + 5e^{-4t}\}$.

$$\mathcal{L}\{3e^{2t} + 5e^{-4t}\} = 3\mathcal{L}\{e^{2t}\} + 5\mathcal{L}\{e^{-4t}\}$$

$$= 3\left(\frac{1}{s-2}\right) + 5\left(\frac{1}{s-(-4)}\right)$$

$$= \boxed{\frac{3}{s-2} + \frac{5}{s+4}}$$

ex: ⑦ Evaluate $\mathcal{L}\{3\cos(2t) + 7\sin(3t)\}$

$$3\mathcal{L}\{\cos(2t)\} + 7\mathcal{L}\{\sin(3t)\}$$

$$= 3\left(\frac{s}{s^2+4}\right) + 7\left(\frac{3}{s^2+9}\right)$$

$$= \boxed{\frac{3s}{s^2+4} + \frac{21}{s^2+9}}$$

ex: ⑧ Evaluate $\mathcal{L}\{\sin(4t)\cos(4t)\}$.

⚠ $\mathcal{L}\{f(t)g(t)\} \neq \mathcal{L}\{f(t)\}\mathcal{L}\{g(t)\}$

Recall that: $\sin(2\theta) = 2\sin\theta\cos\theta$

$$\Rightarrow \mathcal{L}\{\sin(4t)\cos(4t)\} = \mathcal{L}\left\{\frac{\sin(8t)}{2}\right\}$$

$$= \frac{1}{2} \mathcal{L}\{\sin(8t)\}$$

$$= \frac{1}{2} \left(\frac{8}{s^2 + 64} \right)$$

$$= \boxed{\frac{4}{s^2 + 64}}$$

ex: ⑨ Find $\mathcal{L}\{\cos^2 t\}$.

Recall that: $\cos^2 t = \frac{1 + \cos(2t)}{2}$

$$\Rightarrow \mathcal{L}\{\cos^2 t\} = \frac{1}{2} \mathcal{L}\{1\} + \frac{1}{2} \mathcal{L}\{\cos(2t)\}$$

$$= \frac{1}{2} \left(\frac{1}{s} \right) + \frac{1}{2} \left(\frac{s}{s^2 + 4} \right)$$

$$= \boxed{\frac{1}{2s} + \frac{s}{2(s^2 + 4)}}$$

Notations

We will use lowercase letters for functions of t and the corresponding capital letters for their Laplace Transforms.

$$\mathcal{L}\{f(t)\} = F(s)$$

$$\mathcal{L}\{g(t)\} = G(s)$$

$$\mathcal{L}\{y(t)\} = Y(s)$$
